2/10/25

Gradient Operator:

first derivative are implemented using the magnitude of the gradient

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\nabla f = mag(\nabla f) = \left[G_x^2 + G_y^2\right]^{1/2}$$

$$= \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{1/2}$$

$$\nabla f \approx |G_x| + |G_y|$$

Gradient mask (2x2)

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$G_x = (z_8 - z_5)$ & $G_y = (z_6 - z_5)$

$$\nabla f = \left[G_x^2 + G_y^2\right]^{1/2}$$

$$= \left[(z_8 - z_5)^2 + (z_6 - z_5)^2\right]^{1/2}$$

$$\nabla f \approx |z_8 - z_5| + |z_6 - z_5|$$

Roberts cross-gradient operators (2x2)

$G_x = (z_9 - z_5)$ & $G_y = (z_8 - z_6)$

$$\nabla f = \left[G_x^2 + G_y^2\right]^{1/2} = \left[(z_9 - z_5)^2 + (z_8 - z_6)^2\right]^{1/2}$$

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$$

-1	0
0	1

0	-1
1	0

Sobel opertors (3x3)

$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$

$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$

$$\nabla f \approx |G_x| + |G_y|$$

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

* The weight value 2 is to achieve smoothing by giving more importance to the center point. (coefficients)

2-2 frequency domain

* 1-D continuous fourier transform:

$$F(u) = \int_{-\infty}^{\infty} f(x) \cdot e^{-j2\pi ux} \, dx$$

$$f(x) = \int_{-\infty}^{\infty} F(u) \cdot e^{j2\pi ux} \, dx$$

* $F(u)$ of a discrete f^n of one variable, $f(x)$, $x = 0, 1, \ldots M-1$

$$\boxed{F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x) \cdot e^{-j2\pi ux/M}}$$

$$\boxed{f(x) = \sum_{u=0}^{M-1} F(u) \cdot e^{j2\pi ux/M}}$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$\Rightarrow F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x) \cdot \left[\cos 2\pi ux/M - j\sin 2\pi ux/M\right]$$

$$[F(u)] = \left[R^2(u) + I^2(u)\right]^{1/2} \rightarrow \text{magnitude}$$

$$\phi(u) = \tan^{-1}\left(\frac{I(u)}{R(u)}\right) \rightarrow \text{Phase}$$

* Power spectrum

$$P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$

2D discrete Fourier Transform :

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\cdot e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$f(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)\cdot e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$x = 0, 1, 2, \ldots M-1$

$y = 0, 1, 2, \ldots N-1$

$u = 0, 1, 2, \ldots M-1$

$v = 0, 1, 2, \ldots N-1$

$|F(u,v)|$

$$\cdot \frac{1}{M}\sum_{x=0}^{M-1} f(x)\cdot e^{\frac{-j2\pi ux}{m}} \quad (-1)^x$$

$$-1^x = e^{j\pi x}$$

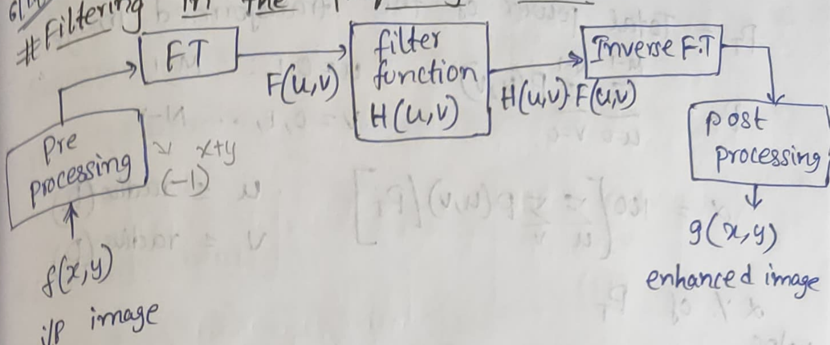
$$\frac{1}{m}\sum_{x=0}^{M-1} f(x)\cdot e^{j\pi x}\cdot e^{\frac{-j2\pi ux}{m}}$$

$$\frac{1}{m}\sum_{x=0}^{M-1} f(x)\cdot e^{-j2\pi x\left(\frac{u}{m}-\frac{1}{2}\right)}$$

$$= \frac{1}{m}\sum_{x=0}^{M-1} f(x)\cdot e^{\frac{-j2\pi x\left(u-\frac{m}{2}\right)}{m}}$$

slides

Filtering in the frequency Domain :



Basic Steps for filtering

1) Multiply the image by $(-1)^{x+y}$ to centre the transform
   ↳ This is pre processing
2) Compute $F(u,v)$, the DFT of the image from (1)
3) Multiply $F(u,v)$ by a filter function $H(u,v)$
4) Compute the inverse DFT of (3)

Filter function

$$H(u,v) = \begin{cases} 0 & \text{if } (u,v) = (M/2, N/2) \\ 1 & \text{otherwise} \end{cases}$$

Ideal low pass filter

$$H(u,v) = \begin{cases} 1, & \text{if } D(u,v) \le D_0 \\ 0, & \text{if } D(u,v) > D_0 \end{cases}$$

Butterworth lowpass filter

$$H(u,v) = \frac{1}{1+\left[D(u,v) + D_0\right]^{2n}}$$

Gaussian filter

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

$D(u,v) \Rightarrow$ Distance from origin

$D_0 \rightarrow$ cutoff freq.

Cutoff freq. of an ideal low pass filter :

P_T = Total power of the transformed image

$$P_T = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} P(u,v) \qquad u = 0,1,\ldots M-1 \quad v = 0,1,\ldots N-1$$

$$\alpha = 100\left[\sum_u \sum_v P(u,v)/P_T\right] \qquad u \le \text{radius}(D_0) \quad v \le \text{radius}(D_0)$$

α % of P_T

7/10/25

TF of highpass filter

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

Ideal highpass filter

$$H(u,v) = \begin{cases} 0, & \text{if } D(u,v) \le D_0 \\ 1, & \text{if } D(u,v) > D_0 \end{cases}$$

Butterworth highpass filter :

$$H(u,v) = \frac{1}{1+[D_0/D(u,v)]^{2n}}$$

$n \Rightarrow$ order of the filter

Gaussian highpass filter :

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

Laplacian in frequency domain :

$$\mathcal{I}\left[\frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}\right] = -(u^2+v^2)F(u,v)$$

$$\Downarrow$$

$$H_1(u,v) = -(u^2+v^2)$$

$\Downarrow$ frequency domain

spatial domain $\Rightarrow \underbrace{g(x,y)}_{\text{enhanced image}} = \underbrace{f(x,y)}_{\text{og image}} - \underbrace{\nabla^2 f(x,y)}_{\text{laplacian o/p}}$

frequency domain $\Rightarrow G(u,v) = F(u,v) + (u^2+v^2)F(u,v)$

new operator $\Rightarrow H_2(u,v) = 1 + (u^2+v^2) = 1 - \underbrace{H_1(u,v)}_{\text{Laplacian}}$

Unsharp masking:

$$f_{hp}(x,y) = f(x,y) - f_{lp}(x,y)$$

Highboost filtering:

$$f_{hb}(x,y) = Af(x,y) -$$

Homomorphic filtering:

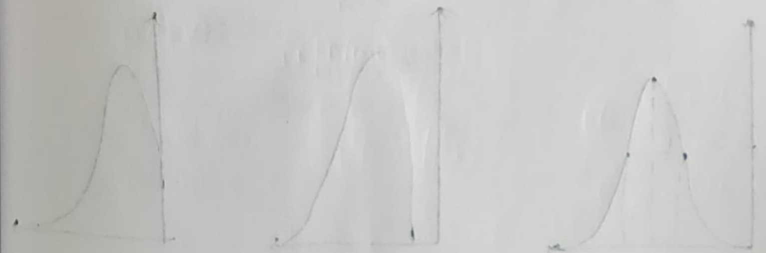
ln: $z(x,y) = \ln f(x,y) = \ln i(x,y) + \ln r(x,y)$

DFT : $Z(u,v) = F_i(u,v) + F_r(u,v)$

$H(u,v)$: $S(u,v) = H(u,v)\,Z(u,v)$

$(DFT)^{-1}$: $s(x,y) = i'(x,y) + r'(x,y)$

exp : $g(x,y) = e^{s(x,y)} = i_0(x,y) \cdot r_0(x,y)$

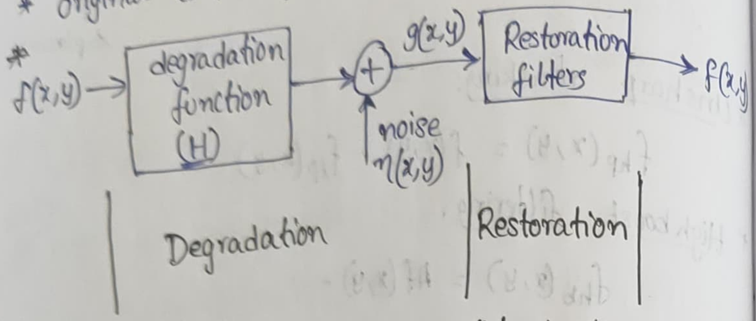


8/10/25

3_1. Image Restoration

* restors a degraded / distorted image to its original content and quality.
* original content quality ≠ good looking

*



model of image degradatⁿ / restoration process

* If H is a linear, position-invariant operator, then the degradation (degraded image) is given in the spatial domain by

$$g(x,y) = (h*f)(x,y) + \eta(x,y)$$

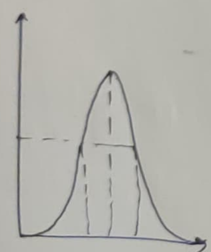
$h(x,y) \rightarrow$ spatial representation of the ~~degradatⁿ~~ degradatⁿ fⁿ.

* In frequency domain

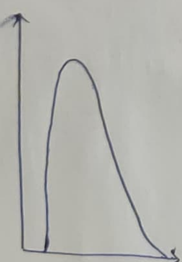
$$G(u,v) = H(u,v)\,F(u,v) + N(u,v)$$

* Sources of noise

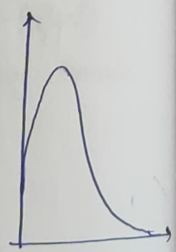
① Gaussian



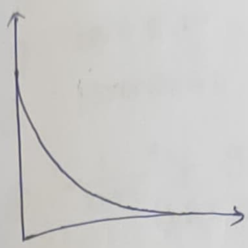
② Rayleigh



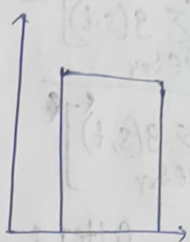
③ Gamma



④ Exponential



⑤ Uniform



⑥ Impulse

Restoration in the presence of Noise only - spatial filtering :

$$g(x,y) = f(x,y) + n(x,y)$$

$$G(u,v) = F(u,v) + N(u,v)$$

Arithmatic Mean Filter

$$f'(x,y) = \frac{1}{mn} \sum_{(s,t)\in S_{xy}} g(s,t)$$

Noise is reduced as a result of blurring.

Geometric Mean Filter

$$f'(x,y) = \left[\prod_{(s,t)\in S_{xy}} g(s,t)\right]^{1/mn}$$

↳ achieves smoothing comparable to the arithematic mean filter

Harmonic Mean Filter

$$f'(x,y) = \frac{mn}{\left[\sum_{(s,t)\in S_{xy}} 1/g(s,t)\right]}$$

13/10/25

Contraharmonic Mean :

$$f'(x,y) = \frac{\left[\sum_{(s,t)\in S_{xy}} g(s,t)\right]^{Q+1}}{\left[\sum_{(s,t)\in S_{xy}} g(s,t)\right]^{Q}}$$

Order Staticstics filter :

- Median filter

$$f'(x,y) = \underset{(s,t)\in S_{xy}}{\text{median}}\{g(s,t)\}$$

- Max and Min filter

$$f'(x,y) = \underset{(s,t)\in S_{xy}}{\max}\{g(s,t)\}$$

$$f'(x,y) = \underset{(s,t)\in S_{xy}}{\min}\{g(s,t)\}$$

- Midpoint filter

$$f'(x,y) = \frac{1}{2} * \left[\underset{(s,t)\in S_{xy}}{\max}\{g(s,t)\} + \underset{(s,t)\in S_{xy}}{\min}\{g(s,t)\}\right]$$

- Alpha-trimmed mean filter

$$f'(x,y) = \frac{1}{(mn-d)} * \sum_{(s,t)\in S_{xy}} g_r(s,t)$$

d ranges from 0 to $mn-1$

$d=0 \Rightarrow f'(x,y) \rightarrow$ arithematic mean filter

$d = \frac{(mn-1)}{2} \Rightarrow f'(x,y) \rightarrow$ median filter

Eg: ① Arithematic mean filter

② Geometric " "

③ Harmonic " "

1	7	5
6	2	3
1	4	2

① $\frac{1}{mn} * \sum_{(s,t)\in S_{xy}} g(s,t) = \frac{1}{9} * (31) = \frac{31}{9} = 3.44 \simeq 3$

② $\left[\prod_{(s,t)\in S_{xy}} g(s,t)\right]^{1/mn} = (1\times7\times5\times6\times2\times3\times1\times4\times2)^{1/9}$

$\simeq 3$

③ $\frac{mn}{\left[\sum_{(s,t)\in S_{xy}} 1/g(s,t)\right]} = \frac{9}{1+\frac{1}{7}+\frac{1}{5}+\frac{1}{6}+\frac{1}{2}+\frac{1}{3}+1+\frac{1}{4}+\frac{1}{2}}$

$= \frac{9}{2+}$

$\simeq 2$

filter for entire image

Eg: $m\times n = 3\times3$

0	0	0	0	0
0	1	3	5	0
0	4	4	3	0
0	5	2	2	0
0	0	0	0	0

① $\frac{1}{9}(0+1+3+4+4)$

$= \frac{12}{9} = 1.33$

$\frac{1}{9}(1+3+5+4+4+3) = \frac{1}{9}(20) = \frac{20}{9} \simeq 2$

$\frac{1}{9}(3+5+4+3) = \frac{15}{9} = \frac{5}{3} \sim 2$

$\frac{1}{9}(4+4+5+2) = \frac{15}{9} = \frac{5}{3} \sim 1$

$\frac{1}{9}(4+4+3+5+2+2) = \frac{20}{9} \sim 2$

$\frac{1}{9}(4+3+2+2) = \frac{11}{9} \sim 1$

$\frac{1}{9}(1+3+4+4+5+2) = 2.11 \sim 2$

$\frac{1}{9}(1+3+5+4+4+3+5+2+2) = \frac{29}{9} \sim 3$

$\frac{1}{9}(3+5+4+3+2+2) \simeq 2$

$\Rightarrow$

1	2	2
2	3	2
2	2	1

AMF

GMF

0	0	0
0	3	0
0	0	0

HMF

0	0	0
0	3	0
0	0	0

Eg: If size is not given, $\Rightarrow$ consider only centre pixel
consider the image and apply mask on it

255	118	112
122	121	111
120	119	112

① AMF $\rightarrow$ 132
② GMF $\rightarrow$ 127
③ HMF $\rightarrow$ 124
④ ContraHarmonic MF $\rightarrow$ Q = -2, Q = 2

④ $$\frac{\left[\sum_{(s,t)\in S_{xy}} g(s,t)\right]^{Q+1}}{\left[\sum_{(s,t)\in S_{xy}} g(s,t)\right]^{Q}} = \frac{\left[255+118+112+122+121+111+120+119+112\right]^{-1}}{(\quad '' \quad)^{-2}}$$

$= 1190$ (for $Q=-2$)

$Q=2 \Rightarrow$ Contra HMF $= 1190$

$Q=-2 \rightarrow 120$
$Q=2 \rightarrow 168$

0.004 + 0.0087 + 0.009 + 0.008 + 0.008 + 0.009 +

14/10/25

Eg: Apply Median, max, Min, Midpoint filter on the given image

1	7	5
6	2	5
1	4	2

Sol: 1 1 2 2 4 5 5 6 7

median $\rightarrow$ 4
max $\rightarrow$ 7
min $\rightarrow$ 1
midpoint $\rightarrow$ $\frac{1}{2}(7+1) = 4$

Adaptive filters

↳ Adaptive Median filter

Z_{min} = min gray level value in S_{xy}
Z_{max} = max gray level value in S_{xy}
Z_{med} = median of gray levels in S_{xy}
Z_{xy} = gray level at (x,y)
S_{max} = max allowed size of S_{xy}

Level A

$A_1 = Z_{med} - Z_{min}$
$A_2 = Z_{med} - Z_{max}$

if $A_1 > 0$ and $A_2 < 0$, go to level B
else increase the window size

Level B

$B_1 = Z_{xy} - Z_{min}$

$B_2 = Z_{xy} - Z_{max}$

if $B_1 > 0$ and $B_2 < 0$, output Z_{xy}

else output Z_{med}

10	20	20
20	15	20
20	25	100

Eg:

$Z_{min} = 10$

$Z_{max} = 100$

$Z_{med} = 10 \quad 15 \quad 20 \quad 20 \quad 20 \quad 20 \quad 20 \quad 25 \quad 100$

$= 20$

Level A

$A_1 = Z_{med} - Z_{min} = 10$

$A_2 = Z_{med} - Z_{max} = 20 - 100 = -80$

Level B

$B_1 = 15 - 10 = 5$

$B_2 = 15 - 100 = -85$

$o/p = Z_{xy} = 15$

Frequency domain filtering for periodic Noise

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \le D(u,v) \le D_0 + \frac{W}{2} \\ 1 & \text{if } D(u,v) > D_0 - \frac{W}{2} \end{cases}$$

27/10/25 : 3.2. Morphological Image Processing

Reflection:

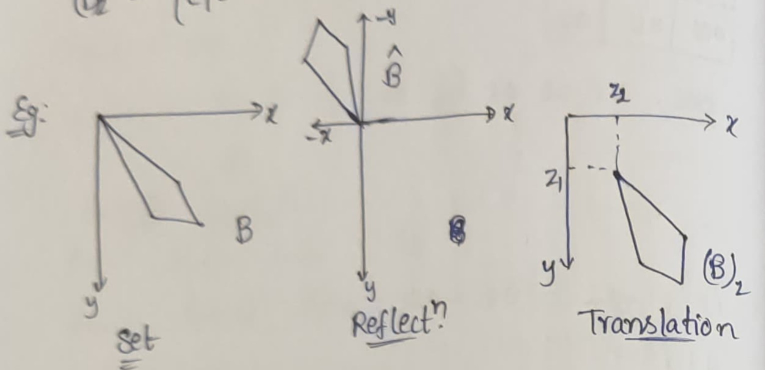
$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$$

Translation:

The translation of a set B by point $z = (z_1, z_2)$ denoted by B_z

$$(B)_z = \{c \mid c = b + z, \text{ for } b \in B\}$$

Eg:



Structure Elements (SE)

small sets or subimages used to probe an image

↳ **Erosion** $A \ominus B$ A → image , B → S.E

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

Also $$A \ominus B = \{z \mid (B)_z \cap A^c = \phi\}$$

28/10/25

Eg: Find the eroded image for the following

1	1	1	1	1	1
1	1	1	0	0	1
1	1	0	0	0	1
1	0	0	0	0	1
1	0	0	0	1	1
1	1	1	1	1	1

OG image
6×6

1	0	1
0	0	0
1	0	1

F.G ← 0

BG ← 1

Structural Element (SE) 3×3

F.G → Fore ground

B.G → Back Ground

eroded image

1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	0	1	1
		0	0		
		1	1		
1	1	1	1	1	1

Eg:

0
0
0
0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
0 0 0 1 1
0
0
0
0
0
0
0
0
0
0
0

Eg:

1	1	1	1
1	1	0	1
0	1	1	1
0	1	0	0

1	1
1	0

0 1 0 0
0 0 0 0
0 1 0 0
0 0 0 0

Dilation $A \oplus B$

$A \to$ image
$B \to$ S.E

Eg:

0	1	0
1	0	0
0	0	0

1	1
0	0

①

0	1	0
1	0	0
0	0	0

②

0	1	1
1	0	0
0	0	0

③

0	1	1	1
1	0	0	0
0	0	0	0

④

0	1	1	1
1	1	0	0
0	0	0	0

0	1	1	1
1	1	1	0
0	0	0	0

⑤

0	1	1	1
1	1	1	1
0	0	0	0

⑥

$$A \oplus B = \{z \mid [(\hat{B})_z \cap A] \subseteq A\}$$

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \phi\}$$

29/10/25

Duality:

Erosion and dilation are duals of each other w.r.t set complementation and reflection.

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

$$(A \oplus B)^c = A^c \ominus \hat{B}$$

$$(A \ominus B)^c = \{z \mid (B)_z \subseteq A\}^c$$

$$= \{z \mid (B)_z \cap A^c = \phi\}^c$$

$$= \{z \mid (B)_z \cap A^c \neq \phi\}$$

$$= A^c \oplus \hat{B}$$

$$(A \oplus B)^c = \{z \mid [(\hat{B})_z \cap A] \subseteq A\}^c$$

$$= \{z \mid [(\hat{B})_z \cap A] \cap A^c = \phi\}^c$$

$$= \{z \mid [(\hat{B})_z \cap A] \cap A^c \neq \phi\}$$

$$= A^c \ominus \hat{B}$$

$$(A \oplus B)^c = \{z \mid (\hat{B})_z \cap A \neq \phi\}^c$$

$$= \{z \mid (\hat{B})_z \cap A^c = \phi\}$$

$$= A^c \ominus \hat{B}$$

* Opening : $A \circ B = (A \ominus B) \oplus B$

Closing : $A \bullet B = (A \oplus B) \ominus B$

Eg: Perform opening and closing on below image using S.E

1	1	1	0	1	1	1
1	1	1	1	1	1	1
1	1	1	0	1	1	1

Image (A)

1	1	1
1	1	1
1	1	1

S.E (B)

Opening : $(A \ominus B) \oplus B$

$A \ominus B$ =

0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0

$(A \ominus B) \oplus B$ =

1	1	1	0	1	1	1
1	1	1	0	1	1	1
1	1	1	0	1	1	1

closing: $(A \oplus B) \ominus B$

$$A \oplus B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}_{5 \times 9}$$

$$(A \oplus B) \ominus B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}_{3 \times 7}$$

3/11/25

Properties of Opening and Closing:

Opening

a) $A \circ B$ is a subset (subimage) of A.

b) If C is subset of D, then $C \circ B$ is a subset of $D \circ B$.

c) $(A \circ B) \circ B = A \circ B$

closing

a) A is a subset (subimage) of $A \bullet B$

b) If C is subset of D, then $C \bullet B$ is a subset of $D \bullet B$.

c) $(A \bullet B) \bullet B = A \bullet B$

Hit-or-Miss transformation : $\odot$

↳ for shape detection

↳ $A \odot B = (A \ominus B_1) \cap (A^c \ominus B_2)$

Eg:

0	1	0
1	1	1
0	1	0

S.E (B)

0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	1	1	1	1	1	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0

input image (A)

$A \ominus B$ =

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

$$A^c = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{matrix} \qquad B = \begin{matrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{matrix}$$

$$B^c = \begin{matrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{matrix}$$

$$\cancel{A^c \ominus B^c} = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$$

$$\therefore \cancel{A \odot B = (A \ominus B) \cap (A^c \ominus B^c)}$$

$$\cancel{= 0 \ 0 \ 0}$$

$$A^c \ominus B^c = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{matrix}$$

$A \odot B = \text{Null}$

Eg:

$$A = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix} \qquad B_1 = \begin{array}{c|c|c} 0 & 0 & 0 \\ \hline 0 & 1 & 0 \\ \hline 0 & 0 & 0 \end{array} \qquad B_2 = \begin{array}{c|c|c} 0 & 1 & 0 \\ \hline 0 & 0 & 0 \\ \hline 0 & 0 & 0 \end{array}$$

$$A \ominus B_1 = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$$

$$A^c = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{matrix} \qquad B_2 = \begin{array}{c|c|c} 0 & 1 & 0 \\ \hline 0 & 0 & 0 \\ \hline 0 & 0 & 0 \end{array}$$

$A^c \ominus B_2 =$

4/11/25

Boundary Extraction

$$\beta(A) = A - (A \ominus B)$$

filet

Region filling / Hole filling

$$X_k = (X_{k-1} \oplus B) \cap A^c \qquad k = 1, 2, 3, \ldots$$

stop iteratⁿ if $X_k = X_{k-1}$

Extraction of Connected Components:

$$X_k = (X_{k-1} \oplus B) \cap A \quad \text{until } X_k = X_{k-1}$$

B: S.E

Convex-Hull:

$$X_k^i = (X_{k-1} \circledast B^i) \cup A \qquad i = 1, 2, 3, 4$$

convex-Hull of A is $C(A) = \bigcup_{i=1}^{4} D^i$

Thinning:

$$A \otimes B = A - (A \circledast B) = A \cap (A \circledast B)^c$$

Thickening

$$A \odot B = A \cup (A \circledast B)$$

$$A \odot \{B\} = ((\ldots(A \odot B^1) \odot B^2) \ldots) \odot B^n)$$

Eg: Hit-miss transformation

$$A = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \boxed{0} & 1 & 1 & 1 & 1 & 1 & \boxed{0} \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$$ Ans

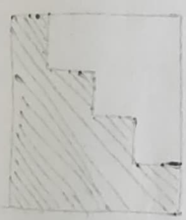
$$B_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad B_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

$$A \ominus B_1 = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & 1 & 1 & & & & \\ & 1 & & & & & \\ & 1 & & & & & \\ & 1 & & & & & \\ & 1 & & & & & \end{matrix}$$

$$A^c = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{matrix} \qquad B_2 = \begin{matrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}$$

$A^c \ominus B_2 =$

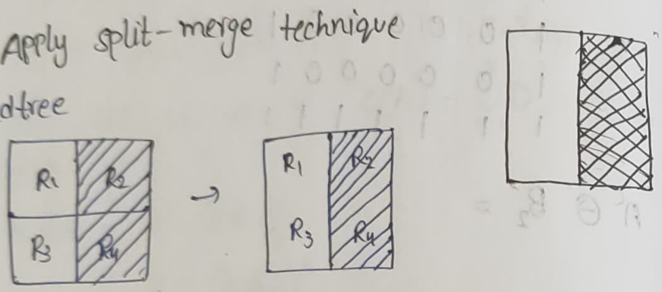


10/11/25

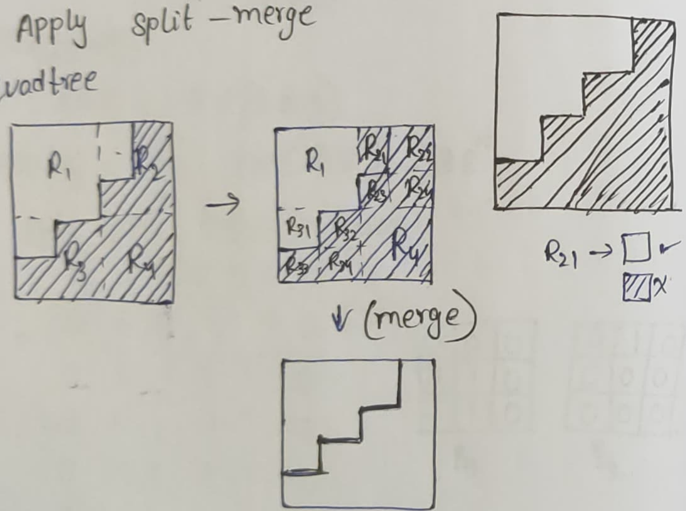
4. Image Segmentation

* segmentat^n algo. are area oriented instead of pixel oriented.
* Segmentat^n is the process of partioning an image into groups of pixels which are homogeneous w.r.t some criteria.
* techniques
  - Global segment^n → segments whole image
  - Local segmentat^n

*Eg: Apply split-merge technique quadtree



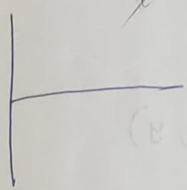
Eg: Apply split-merge quadtree

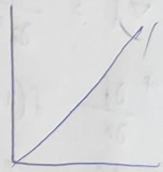


Cluster Distance Measures

1) single link : $d(C_i, C_j) = \min\{d(x_{ip}, x_{jq})\}$
2) complete link : $d(C_i, C_j) = \max\{d(x_{ip}, x_{jq})\}$
3) Average : $d(C_i, C_j) = \text{avg}\{d(x_{ip}, x_{jq})\}$

i) step edge ii) Line edge iii) Ramp edge





12/11/25

Gradient operator

$$\nabla f = \begin{vmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{vmatrix}$$

* gradient orientation

$$\phi(\nabla f) = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

* Edge strength = magnitude of the gradient
  Edge direction = angle " " "
* Roberts cross gradient operators →

-1	0
0	1

0	-1
1	0

Prewitt operators →

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

Sobel →

17/11/25

$$\nabla \cdot \nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

$\nabla \cdot \nabla = \nabla^2 \rightarrow$ Laplacian operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

* $\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$
* $\frac{\partial^2 f}{\partial x^2} = f(x+1, y) - 2f(x, y) + f(x-1, y)$
* $\frac{\partial^2 f}{\partial x^2} = f(x, y+1) - 2f(x, y) + f(x, y-1)$

Drawbacks of the Laplacian Operator:

→ Useful directional info. is not available by the use of a laplacian operator

→ The laplacian, being an approximatⁿ to the second derivative, doubly enhances any noise in the image.

Laplacian of Gaussian (LOG):

↳ Smoothens the image through convolution with a gaussian - shaped kernel followed by applying the laplacian operator.

* $h(r) = -e^{\frac{-r^2}{2\sigma^2}}$ where $r^2 = x^2 + y^2$

σ : the standard deviation

Laplacian of h is

$$\nabla^2 h(r) = -\left[\frac{r^2 - \sigma^2}{\sigma^4}\right] e^{\frac{-r^2}{2\sigma^2}}$$

LOG mask

0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

Eg: $A(x, y) = (1, 2)$

$B(x, y) = (3, 4)$

$y = ax + b$

$y - 2 = 1(x - 1)$

$y = x + 1$

$y = mx + c$

$\Rightarrow c = -mx + y$

$\Rightarrow \boxed{c = -m + 2}$ (1,2)

$c = 0$	$m = 2$ ✓
$c = 1$	$m = 1$
$c = 2$	$m = 0$ ✓

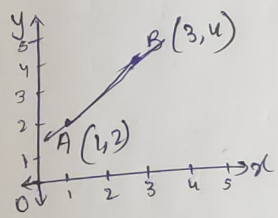
(2,2)

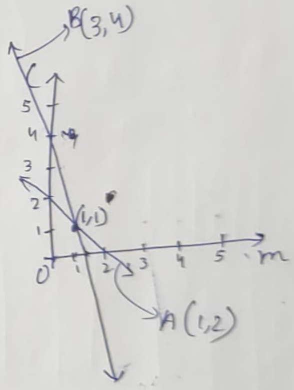
$c = -mx + y$

$c = -3m + 4$ (3,4)

$c = 0$ $m = \frac{4}{3}$ (1.33, 4)

$m = 0$ $c = 4$





• $(m, c) = (1, 1)$

$y = mx + c$

$(x, y) = (1, 2)$

$(x, y) = (3, 4)$